The Journal of the Acoustical Society of America

MAY 01 2026

A few-shot learning method for underwater acoustic target recognition based on generative data augmentation[a)]

Wei Huang ; Qirui Zhang; Bowen Zhao; Tianhe Xu; Hao Zhang

Check for updates

J. Acoust. Soc. Am. 159, 3773–3791 (2026)
https://doi.org/10.1121/10.0043727

View Online
Export Citation

Articles You May Be Interested In

Knowledge-inspired and sample-generation-based spectral augmentation for few-shot underwater acoustic target recognition

J. Acoust. Soc. Am. (July 2026)

10 October 2026 10:37:30

ASA
LEARN MORE
Advance your science and career as a member of the
Acoustical Society of America

ASA
ACOUSTICAL SOCIETY OF AMERICA

JASA ARTICLE

A few-shot learning method for underwater acoustic target recognition based on generative data augmentation[a)]

Wei Huang,[1] Qirui Zhang,[1] Bowen Zhao,[1] Tianhe Xu,[2] and Hao Zhang[1,b)]
[1]Faculty of Information Science and Engineering, Ocean University of China, Qingdao, Shandong 266100, China
[2]School of Space Science and Engineering, Shandong University at Weihai, Weihai, Shandong 264200, China

ABSTRACT:
The recognition of underwater acoustic targets (UATR) is of great significance for the protection of marine diversity and national defense security. The development of deep learning provides new opportunities for UATR, but faces challenges brought by the scarcity of reference samples and complex environmental interference. To tackle this problem, we propose a generative discriminative collaborative framework, a variational auto-encoder boosted learning framework based on latent space completion. Rooted in the core contradiction arising from the incompleteness of intra-class manifolds and the instability of discriminative boundaries, this framework incorporates the premise of latent space continuity. Leveraging a structure-preserving generative reconstruction mechanism, it implicitly supplements the original dataset, which in turn enables the reconstruction of intra-class distributions that are more continuous, integral, and discriminative at the feature level. In this paper, we construct a three-stage pipeline system consisting of auto-clean cut unified preprocessing, latent reconstruction variational auto-encoder multi-scale latent space reconstruction, and an acoustic identification model. Furthermore, by establishing a staged modeling workflow, data purification, latent space completion, and discriminative optimization converge on their individual objectives independently while maintaining overall synergy, thus forging a robust recognition paradigm tailored to few-shot learning scenarios. © *2026 Acoustical Society of America*.
https://doi.org/10.1121/10.0043727

(Received 4 December 2025; revised 10 March 2026; accepted 7 April 2026; published online 1 May 2026)
[Editor: Haiqiang Niu] Pages: 3773–3791

10 October 2026 10:37:30

I. INTRODUCTION

Sound waves are the only known form of energy that can propagate over long distances underwater and are therefore widely regarded as the best information carrier to detect and identify underwater targets.[1] The recognition of underwater acoustic targets (UATR) is of vital importance in the exploration of marine resources, the conservation of marine biodiversity, and national defense security. In terms of biological conservation, underwater acoustic recognition technology can support population monitoring and behavioral analysis of marine mammals such as whales and dolphins, avoiding interference from human activities on their habitats. At the same time, it can monitor the acoustic signals of illegal fishing vessels and help protect fish resources. In emergency rescue, UATR can provide rapid ship-type identification to assist accident response and situational awareness, shorten search and rescue time, and thus improve the rescue success rate.

In recent years, the rapid development of theoretical research in deep learning has provided novel solutions for research directions in UATR.[2] However, the complex underwater environment still poses serious challenges that affect the accuracy performance.[3] The marine environment is permeated with noise from diverse sources, including biological activity, maritime shipping, and natural turbulence. This ambient noise can obscure weak radiated acoustic signals emitted by the target, disrupt the detection of acoustic features, diminish the precision of feature extraction, and ultimately compromise the reliability of UATR recognition. In addition, there are strong multipath effects during underwater signal propagation, resulting in selective frequency fading of the received target signal, which brings difficulties in accurately recognizing the target. Finally, the expensive economic cost and the difficult maintenance conditions of underwater equipment result in a scarcity of data samples, posing challenges to fully study the characteristics of underwater target signals.

Conventional audio processing approaches, using time-domain analysis,[4] frequency-domain analysis,[5] and cepstral analysis,[6] have been extensively employed in underwater acoustic research. Nevertheless, these approaches face intrinsic technical limitations when dealing with complex underwater acoustic environments. Time-domain analysis is fundamental but shows significant limitations in characterizing non-stationary signals. Frequency-domain analysis techniques, particularly those based on Fourier transforms, face significant limitations stemming from their inherent assumption of stationary signals. The relative motion between underwater sound sources and receivers gives rise to

[a)]This paper is part of a special issue on Generative and Physics-Informed Artificial Intelligence for Acoustics.
[b)]Email: zhanghao@ouc.edu.cn.

 © 2026 Acoustical Society of America

Doppler frequency shifts that can reach 2%–3% of the central frequency, which in turn induces spectral smearing and consequent degradation in signal resolution. Moreover, the intricate nature of multipath propagation within underwater acoustic channels causes frequency-selective fading effects. In such scenarios, the depth of the signal notches may exceed 15 dB, severely distorting the spectral profiles that are crucial for accurate target discrimination. The frequency cepstral analysis approach, with the Mel frequency cepstral coefficient (MFCC) as a prominent representative, demonstrates the efficacy in emulating human auditory perception. However, it faces notable adaptability issues when applied in underwater scenarios. Specifically, the logarithmic compression step involved in MFCC computation tends to amplify the impact of low-frequency ambient noise, a phenomenon that can potentially obscure the faint spectral components of target signals. To address these limitations, Abdul and Al-Talabani[7] reported an improved performance for automatic speech recognition by combining MFCCs with deep learning architectures, but the method continues to be susceptible to the effects of multipath underwater distortion. Tang *et al.*[8] developed a unique three-dimensional spectrogram representation processed through 3DSNet, which independently handles the temporal and frequency domains. Although the method improves feature representation capabilities, it still struggles with the significant temporal variability of underwater biological sounds, especially when dealing with intermittent vocalizations.

In addition to designing features that are more suitable for UATR recognition, the progress of underwater acoustics research has witnessed the advent of various machine learning-based audio classification algorithms. Traditional early methods relied mainly on statistical models, particularly the Hidden Markov model (HMM) for temporal sequence analysis and the Gaussian Mixture model (GMM) for characterizing the distribution of acoustic features. Although these methods established important groundwork, they encounter specific technical constraints in underwater settings. By combining Gamma-Tone Cepstral Coefficients with a 20-state HMM, Mohammed *et al.*[9] developed a classifier, which was verified using field data. However, in practical use, HMMs have three key limitations: the Viterbi algorithm for state sequence estimation has $O(N^2T)$ computational complexity, causing scalability issues for long recordings, their sensitivity to state initialization often leads to suboptimal local maxima in Baum-Welch training, and critically, the Markov state independence assumption fails to capture long-range contextual dependencies in marine signals. Unlike GMMs, they exhibit superior flexibility through multi-modal distribution modeling, which is illustrated by the work of Xue and Jiang[10] on adaptive UWB signal classification under different meteorological conditions. Due to its ability to capture complex spectral structures through weighted Gaussian components, GMM is applicable to diverse acoustic features of underwater targets, but suffers from inherent constraints: the convergence time of the expectation-maximization algorithm increases in a cubic manner with the number of mixed components, and insufficient Gaussian components can lead to poor adaptation to complex spectral time patterns.

Bist and Singh[11] systematically examined support vector machine (SVM) optimization methods, emphasizing the challenges of model selection and computational efficiency. SVMs, while effective in constructing optimal hyperplanes for binary classification, face difficulties in kernel selection and high complexity when applied to underwater acoustics. Decision trees process multidimensional acoustic data through hierarchical segmentation, but are easily affected by impulse noise, and local segmentation criteria cannot capture global mode boundaries. The random forest improves robustness through ensemble learning, but remains challenged by the exponential growth of training time with high-dimensional features and performance degradation with class-imbalanced marine data. With the evolution of statistical models in underwater audio classification, the inherent limitations of these models have become increasingly prominent: Computational efficiency tends to decline in high-dimensional feature spaces, and traditional methods show an exponential increase in computational costs when feature dimensions exceed 50.

In recent years, the emergence of deep learning technology has brought new possibilities for underwater audio classification, leading to a transformative change in this field. Convolutional neural networks (CNNs) and recurrent neural networks (RNNs) are two dominant architectural paradigms used for UATR. Aodha *et al.*[12] have developed an open-source system that leverages a supervised deep convolutional neural network (BatDetect CNN), which allows direct feature learning from raw audio data to automate the detection of bat echolocation calls in noisy recordings in the real-world. Nevertheless, this approach has inherent limitations in generalizing to impulsive acoustic signals amid high-noise environments, and its robustness against non-stationary background noise still remains to be enhanced. Yang *et al.*[13] put forward an auditory perception-inspired deep convolutional neural network (ADCNN) for UATR, which performs the integrated processing of decomposition, feature extraction, and classification for ship-radiated noise. Despite its architectural innovation, the model showed limited ability to suppress signal reverberation induced by multipath propagation, and its compensation mechanism for Doppler shifts generated by high-speed moving targets remained incomplete. Miao *et al.*[14] developed a time-frequency feature network (TFFNet) for underwater acoustic signal classification, which integrates an effective feature pyramid (EFP) forward fusion strategy to reduce network complexity. Although the network exhibited high efficiency in noisy environments, it still had inherent limitations in the feature extraction resolution for low-frequency-dominant underwater acoustic signals, and its modeling capability for dispersion effects in cross-water propagation remained weak.

To improve feature capture capability, Xiao *et al.*[15] proposed an attention-based neural network (ABNN) for multi-

10 October 2026 10:37:30

https://doi.org/10.1121/10.0043727

source interference pressure spectrograms. Although the attention mechanism enhanced focus on critical features, its effectiveness in suppressing underwater-specific coherent interference remained inadequate. Xue *et al.*[16] proposed the CamResNet model by incorporating a channel attention mechanism into ResNet, effectively improving the robustness to Doppler shifts, and Feng and Zhu[17] proposed a transformer-based UATR-Transformer model, demonstrating strong performance in complex marine environments. However, these models require a large number of training samples and are difficult to meet in practical underwater applications.

Current data-driven methodologies lean heavily on large-scale labeled datasets for their implementation, frequently failing to account for the practical hurdles associated with obtaining extensive data samples in real-world marine settings. Although these approaches deliver outstanding performance under optimal conditions, they suffer from notable limitations when faced with the few-shot learning scenarios that are prevalent in UATR. To address these constraints, some researchers have begun exploring multi-task learning (MTL) frameworks. Thanda and Venkatesan[18] developed an audio-visual speech recognition system (MTL-DNN) based on MTL, which significantly improves robustness in high-noise environments through simultaneous learning of audio-visual fusion features and visual feature mapping. Kong *et al.*[19] proposed a cross-task learning framework integrating audio tagging, sound event detection, and spatial localization, establishing a unified baseline for multi-task audio recognition in the DCASE 2019 Challenge. Khandelwal and Das[20] designed an MTK-based acoustic scene classification (ASC) method that employs topic-based soft labels and a mutual attention mechanism to optimize the representation of characteristics in diverse acoustic environments. Although MTK provides a model training method for dealing with small sample sizes, negative transfer problems are prone to occur when the sample distribution is uneven.

In order to address the few-shot learning problem with an imbalanced distribution, scholars have proposed some generative models to enhance the sample data and weaken the overfitting phenomenon of the model. Liu *et al.*[21] expanded the data set using correction strategies in the time-domain and in the frequency-domain based on generative adversarial networks (GANs) to improve model generalization. In an attempt to solve the problem of few-shot learning and imbalanced categories of underwater target data, Jiang *et al.*[22] proposed an improved DCGAN model to generate high-quality data to expand the training set, combined with S-ResNet for classification. Yao *et al.*[23] proposed a UATR framework based on DCGAN and residual CNN, which converted the original signal into MFCC features, generated enhanced samples using DCGAN, and classified them using ResNet18.[24] However, these GAN-based models suffer from training instability and pattern collapse issues.

To tackle this problem, we propose a generative discriminative collaborative framework variational auto-encoder boosted learning framework (VAE-BL) based on latent space completion. This framework starts from the fundamental contradiction between incomplete intra-class manifolds and unstable discriminative boundaries, introduces the assumption of latent space continuity, and implicitly supplements the original data through a structure-preserving generative reconstruction mechanism, thereby reconstructing a more continuous, complete, and discriminative intra-class distribution at the feature level. To achieve this goal, we construct a three-stage pipeline system consisting of auto-clean cut (ACcut) unified preprocessing, latent reconstruction variational auto-encoder (LR-VAE) multi-scale latent space reconstruction, and acoustic identification model (AIM). By constructing a staged modeling process, data purification, latent space completion, and discrimination optimization converge independently on their respective goals and maintain synergy as a whole, thus forming a robust recognition path for short-shot learning. The main contributions of this study are summarized as follows:

(1) To address the issue of overfitting in few-shot object recognition, we propose a generative enhancement framework for UATR, and construct a multi-level modeling path of "preprocessing latent space generation expansion discriminative recognition." This framework effectively expands the distribution of intra-class data through latent space reconstruction, significantly improving the learnability and stability of the model under small sample conditions.
(2) We design an LR-VAE generation model with multi-scale skip connections to enhance the continuity and separability of latent space expression by combining shallow local acoustic information with deep high-level acoustic features in a symmetrical structure. Simultaneously combining frozen BEATs features with a lightweight LightGBM classifier, a recognition framework that combines high accuracy and efficiency is constructed.
(3) To evaluate the performance of the proposed model, we tested it on two popular real-measured datasets, and its performance exceeded that of recent baseline models.

The remainder of this paper is organized as follows. The system model is proposed in Sec. II. The results and discussions are given in Sec. III, and the conclusions are drawn in Sec. IV.

II. SYSTEM MODEL

In this section, we provide a detailed description of the proposed VAE-BL few-shot learning underwater acoustic identification framework, including its overall architecture, core functional modules, and the processing logic that integrates them. The framework is also conceptually contrasted with conventional large-sample acoustic fingerprinting systems to highlight its suitability under data-scarce and noise-varying marine environments.

10 October 2026 10:37:30

A. Overall framework

Underwater acoustic identification tasks often encounter a pronounced data scarcity, which limits the ability of end-to-end models to learn stable and discriminative representations. To mitigate these constraints, we propose the VAE-BL, a few-shot learning recognition framework designed for raw waveform input and constructed around a three-stage modeling pipeline comprising data preprocessing, generative enhancement, and discriminative classification. The complete architecture is illustrated in Fig. 1.

Due to notable variability in sampling rates, noise conditions, and energy distributions across raw underwater recordings, directly feeding unprocessed waveforms into feature extractors may lead to unstable input distributions and training instability. Therefore, prior to entering the generative and discriminative stages, each signal undergoes systematic preprocessing involving resampling, sliding-window segmentation, energy-based validity assessment, and amplitude normalization.

To ensure consistency of input data, we propose the ACcut module as a data standardization tool. ACcut performs unified resampling, fixed-window slicing, and energy-validity evaluation to exclude silent or highly unstable segments, producing five-second waveform clips with consistent duration and perceptual quality. The purpose of ACcut is not feature enhancement but rather the establishment of a well-conditioned and homogeneous input distribution suitable for stable generative modeling.

In few-shot learning ship-radiated noise identification, the limited availability of real data may result in insufficient coverage of natural intra-class variations caused by diverse recording conditions and acoustic environments. This sparsity typically results in dispersed intra-class distributions and blurred inter-class boundaries, thereby posing challenges for classifier generalization. To address these issues, we introduce LR-VAE, a structure-preserving generative model that leverages latent-space continuity to expand intra-class distributions. LR-VAE operates on a unified 128×128 log-Mel representation, encodes it into a 1024-dimensional latent vector, and reconstructs its spectral structure through reparameterized sampling and mirrored decoding. The Griffin–Lim algorithm[25] is subsequently applied to recover audible waveforms, enabling structure-preserving enhancement that tends to produce more compact intra-class clusters and clearer inter-class boundaries.

Finally, the AIM performs discriminatory classification. The reconstructed waveforms, along with the original segments, are fed into a frozen BEATs[26] acoustic encoder to extract 768-dimensional global embeddings, which are then classified using a LightGBM[27] model. By integrating ACcut-based data purification, LR-VAE-based generative expansion, and lightweight classification based on AIM, the VAE-BL framework provides a robust and scalable approach to underwater acoustic identification under limited-sample and noise-varying conditions. The modules are loosely coupled and incur a low training cost, making the framework suitable for deployment in resource-constrained environments.

B. Data preprocessing and standardization

The ACcut module provides consistent and reliable data entry for the entire framework. It outputs 5-s standardized audio clips with consistent duration and perceptual quality. The primary goal of ACcut is not to introduce feature enhancement, but to establish a well-conditioned and model-friendly input distribution for downstream generative modeling. There are mainly four steps in the ACcut module.

1. Resampling

Given a continuous-time waveform $x(t)$, its discretized version sampled at an original sampling rate f_0 is expressed as $\boldsymbol{x}_n = x(n/f_0)$, where $\boldsymbol{x}_n$ denotes the discrete-time sample sequence. To obtain a unified spectral resolution, the waveform is resampled at a target sampling rate f_r,

$$\boldsymbol{x}_r[n] = x\left(\frac{n}{f_r}\right), \quad f_r = 16\,\text{kHz}. \tag{1}$$

This target sampling rate is consistently adopted throughout the processing pipeline, including log-Mel spectrogram construction, LR-VAE generative modeling, and all experimental evaluations reported in Sec. III.

2. Global energy baseline estimation

To evaluate whether a segment cut contains valid acoustic information, the average energy of the resampled signal E_{avg} is computed as

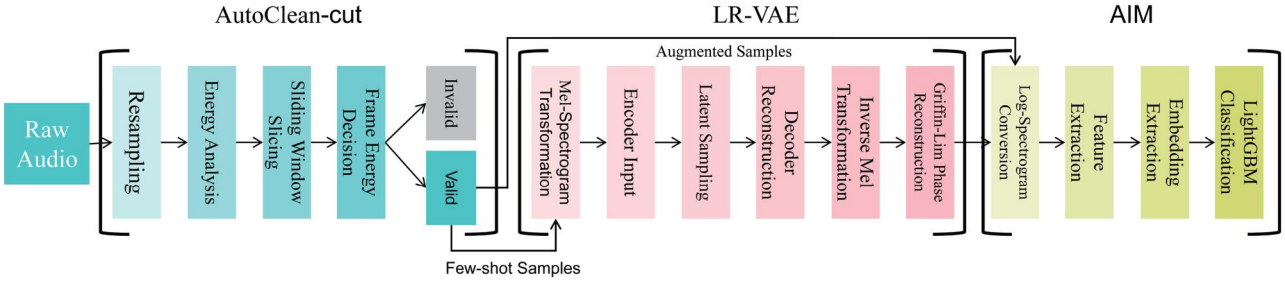


FIG. 1. Overall architecture of the proposed VAE-BL framework.

10 October 2026 10:37:30

https://doi.org/10.1121/10.0043727

$$E_{\text{avg}} = \frac{1}{N}\sum_{n=0}^{N-1} \boldsymbol{x}_r^2[n], \tag{2}$$

where N is the total number of samples. For each set of experiments, E_{avg} is estimated exclusively from the training data. The resulting statistic is treated as a fixed global baseline and is consistently applied to the validation and test datasets.

3. Sliding-window segmentation

A fixed-window sliding strategy is adopted to match the input size required by the downstream components. With window length $T_w = 5$ s and hop length $T_h = 1$ s, the corresponding numbers of waveform samples are calculated as $L = T_w f_r, H = T_h f_r$, where L denotes the number of samples contained in each 5 s analysis window, and H represents the number of samples shifted between two adjacent windows. Assuming there are a total of $K = \lfloor (N-L)/H \rfloor + 1$ windows extracted, the kth sample slice will be

$$\boldsymbol{x}_k[n] = \boldsymbol{x}_r[n + kH], \quad 0 \le n < L. \tag{3}$$

Approximately 80% overlap is maintained via $T_h = 0.2T_w$. This segmentation ensures not only an increased number of effective samples but also a preserved temporal continuity, which is beneficial for downstream generative modeling.

4. Segment energy computation and adaptive threshold setting

To adaptively distinguish between effective acoustic frames and low-energy background noise, a global energy baseline is estimated from the resampled signal. For each set of experiments, all energy statistics used for threshold estimation are computed exclusively from the training data. The resulting statistics are treated as fixed global baselines and are consistently applied to the validation and test datasets without re-estimation. To account for varying noise conditions across different recording environments, the global baseline energy E_k is obtained by computing the mean energy over all resampled clips,

$$E_k = \frac{1}{N}\sum_{n=0}^{N-1} \boldsymbol{x}_k^2[n]. \tag{4}$$

Based on the global baseline, an adaptive energy threshold η is defined as

$$\eta = E_{\text{avg}} - 2\sigma(E_{\text{avg}}), \tag{5}$$

where $\sigma(E)$ represents the empirical standard deviation of the energy distribution computed over the training set clips only. A frame is considered valid if its short-term energy satisfies $E_k > \eta$. Frames with $E_k \le \eta$ are regarded as silent or pronounced contaminated segments and are therefore discarded. This adaptive mechanism enables reliable removal of low-energy noise, silent intervals, and unstable frames while retaining the characteristic components of the underwater acoustic signal. The resulting segments form a clean and consistent input basis for subsequent LR-VAE generative modeling.

5. Amplitude normalization

After the ACcut filtering process, the remaining valid segments exhibit consistent durations but may still contain residual amplitude variations. To further eliminate such inconsistencies, each segment $\boldsymbol{x}[n]$ is normalized as

$$\widetilde{\boldsymbol{x}}_k[n] = \frac{\boldsymbol{x}_k[n]}{\max\left(|\boldsymbol{x}_k[n]|\right) + \varepsilon}, \tag{6}$$

where ε is a small constant added to prevent numerical instability. This normalization confines all sample amplitudes to the range $[-1, 1]$, ensuring stable behavior during subsequent frequency-domain transformation.

To maintain consistent input dimensions, each valid 5-s waveform segment is converted to a 128×128 log-Mel spectrogram for further processing. After this procedure, the cleaned waveform is treated as zero-mean, free from abnormal low-energy or silent intervals, and serves as a well-conditioned training sample for the LR-VAE module.

C. LR-VAE model

VAEs are widely used generative models that learn probabilistic latent representations by characterizing the joint distribution between observed data and latent variables. While standard VAE formulations provide a principled mechanism for learning compact generative embeddings, their typical architectures, often relying on fully connected layers or shallow convolutional blocks without feature-level skip connections, are not well suited for underwater acoustic signals, which commonly exhibit strong background noise, limited sample diversity, and substantial intra-class variability.

To obtain more robust latent representations under few-shot learning conditions and to preserve fine-grained spectral structure during reconstruction, we develop a latent reconstruction model termed LR-VAE. Unlike conventional VAE designs, LR-VAE incorporates a convolutional encoder–decoder backbone with multi-scale skip-concatenation pathways. These pathways allow local time–frequency cues to be propagated across symmetric encoder and decoder stages, while the latent bottleneck captures compact global structures. This design improves reconstruction fidelity and helps maintain discriminative patterns that are relevant for downstream classification.

The LR-VAE module consists of three components: Mel-spectrogram construction, latent-space encoding and decoding, and waveform reconstruction. The encoder ingests 5-s waveform segments processed by ACcut, and the reconstructed outputs provide both enhanced Mel-spectrograms and time-domain signals to support classification in the following AIM module.

10 October 2026 10:37:30

1. Mel-spectrogram construction

Given an ACcut-processed waveform $\boldsymbol{x}_r[n]$, the short-time Fourier transform (STFT) is first applied to obtain a time–frequency representation. For the kth frame and mth frequency bin, the STFT is defined as

$$X(m,k) = \sum_{n=0}^{N_w-1} \boldsymbol{x}_r[n + kH_w]\, w[n] e^{-j2\pi nm/N_w}, \tag{7}$$

where $N_w = 1024$ is the length of the analysis window, $H_w = 512$ is the hop size, and $w[n]$ denotes the Hann window. The magnitude spectrogram is then obtained as

$$P(m,k) = |X(m,k)|. \tag{8}$$

The magnitude spectrogram is passed through a 128-channel Mel-filterbank. For the ith Mel filter, the output is computed as

$$E(i,m) = \sum_{k=0}^{L-1} P(m,k) H_i(k), \tag{9}$$

where $H_i(k)$ denotes the triangular Mel filter. A log transformation is then applied to suppress dynamic-range variation,

$$M(i,m) = \log\left(E(i,m) + \varepsilon\right), \tag{10}$$

where ε ensures numerical stability.

To avoid numerical underflow during logarithmic operations, a small constant δ is introduced. Since the overall energy distribution may vary between time–frequency segments, each frame is independently normalized using a min–max scheme. The normalized log-Mel spectrogram $\widetilde{M}(i,m)$ is defined as

$$\widetilde{M}(i,m) = \frac{M(i,m) - M_{\min}}{M_{\max} - M_{\min} + \delta}, \quad 0 \le \widetilde{M}(i,m) \le 1, \tag{11}$$

where $M_{\min}$ and $M_{\max}$ denote the global minimum and maximum logarithmic magnitude values across all Mel filters and all frames within the current segment, and δ is a small positive constant used to stabilize the denominator.

To obtain a fixed-dimensional input representation while preserving the temporal information of the entire 5-s waveform segment, the time axis of the log-Mel spectrogram is normalized to 128 frames. When the number of STFT frames is less than 128, zero padding is applied to supplement the deficit. When it exceeds 128, frames are uniformly sampled along the time axis to form 128 representative frames that are distributed across the full duration of the segment, rather than retaining only the initial portion. Notably, this operation is not equivalent to recomputing the STFT with an excessively large frame shift, nor does it discard a contiguous temporal portion of the signal. Instead, the complete 5-s waveform is first transformed into a full-resolution time–frequency representation, and the subsequent temporal normalization is performed only on the constructed log-Mel spectrogram. Therefore, the resulting 128-frame representation still covers the full temporal extent of the original segment and preserves its global temporal structure.

The underwater ship-radiated noise considered in this work is dominated by relatively stable spectral envelopes and long-duration tonal or broadband components, whose discriminative cues mainly lie in the overall spectral distribution and its temporal evolution across the entire segment rather than in highly localized frame-level fluctuations. Under this condition, uniformly sampling representative frames from the entire segment can retain the dominant class-related acoustic characteristics while reducing redundant local temporal correlations in a controlled manner. Moreover, adopting a fixed 128×128 representation provides a unified input format for the LR-VAE model, enabling consistent feature reconstruction and stable model training across different recordings. Therefore, the temporal normalization used here should be understood as a compact global representation of the full 5-s segment rather than a simple compression that removes essential task-relevant information.

It is also worth noting that this normalization strategy is not restricted to a specific number of STFT frames. If variations in signal duration or preprocessing settings result in a different number of frames, the same normalization principle can still be applied: frames are zero-padded when the frame number is less than 128, and uniformly sampled when it exceeds 128. Therefore, the preprocessing pipeline remains robust to variations in the raw input dimension and does not affect the applicability of the subsequent LR-VAE modeling framework. This procedure yields a unified 128×128 normalized log-Mel matrix $\widetilde{\boldsymbol{M}}$, which is then used as input to the LR-VAE.

2. Skip-connected latent reconstruction

The LR-VAE employs a convolutional encoder–decoder architecture with U-Net-inspired skip connections. These connections allow the decoder to leverage encoder features at matched resolutions, mitigating information loss due to downsampling and supporting consistent reconstruction of time–frequency details.

a. Convolutional encoder. The encoder accepts a normalized log-Mel input $\boldsymbol{M} \in \mathbb{R}^{128\times128}$ and processes it through a sequence of 2D convolutional blocks. Through hierarchical downsampling, the encoder compresses the input into progressively lower resolutions while maintaining informative spectral cues. The detailed architecture is given in Fig. 2.

In Fig. 2, The left branch consists of a four-stage downsampling convolutional encoder (Enc1–Enc4), which progressively compresses the input Mel-spectrogram and extracts multi-scale spectral features. The middle block corresponds to the latent module, including a flattening operation, the two fully connected layers that produce the mean vector $\boldsymbol{\mu}$ and the log-variance vector $\log \boldsymbol{\sigma}^2$, and the

10 October 2026 10:37:30

https://doi.org/10.1121/10.0043727

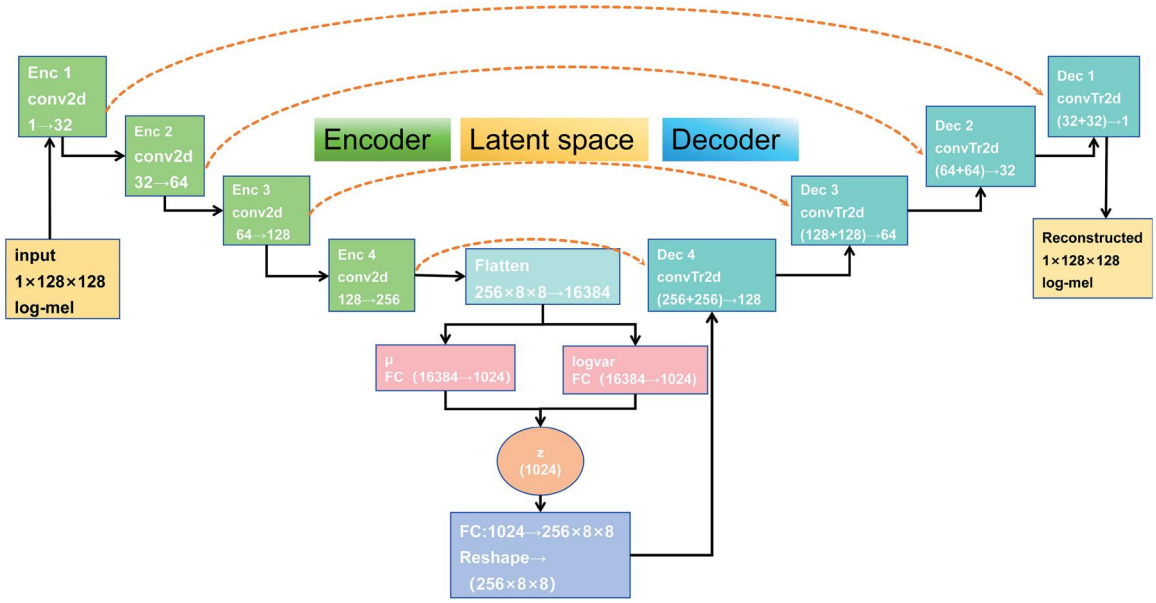


FIG. 2. Structure of the convolutional encoder in the proposed LR-VAE.

reparameterized latent variable z obtained by stochastic sampling. The right branch contains four transposed-convolution decoder stages (Dec1–Dec4) that reconstruct the latent representation back to a two-dimensional time–frequency structure with resolution comparable to the input.

To preserve fine-grained spectral information, four cross-layer skip connections are established between the encoder and decoder stages. These skip pathways concatenate feature maps of matching spatial resolutions, allowing shallow structural cues to propagate to deeper layers and thereby enhancing reconstruction fidelity. Skip connections are integrated between the encoder and decoder at four hierarchical resolution levels, corresponding to relative downsampling factors of $\{16, 8, 4, 2\}$ with respect to the input spectrogram. These skip pathways enable the decoder to recover fine-grained spectral details that may be lost during progressive downsampling, while simultaneously facilitating stable gradient propagation across the network. For clarity, a notation of the form $\mathrm{Conv2d}\, a \to b$ denotes a convolutional layer that maps a input channels to b output channels, whereas $c \times c \to d \times d$ indicates the corresponding spatial resolution change of the feature map. The final encoder-stage feature map is denoted by $H_{\mathrm{enc}}^{(4)} \in \mathbb{R}^{256\times8\times8}$. This feature map is then flattened into a latent vector

$$h = \mathrm{flatten}\left(H_{\mathrm{enc}}^{(4)}\right) \in \mathbb{R}^{16384}. \tag{12}$$

b. Latent variable modeling. Two fully connected layers estimate the mean and logarithmic variance of the latent distribution: $\boldsymbol{\mu} = \boldsymbol{W}_\mu \boldsymbol{h} + \boldsymbol{b}_\mu$, $\log \boldsymbol{\sigma}^2 = \boldsymbol{W}_\sigma \boldsymbol{h} + \boldsymbol{b}_\sigma$, where the parameters $(\boldsymbol{W}_\mu, \boldsymbol{W}_\sigma, \boldsymbol{b}_\mu, \boldsymbol{b}_\sigma)$ are the parameters learned during training.

The latent variable z is sampled using the reparameterization trick,

$$z = \boldsymbol{\mu} + \exp\left(0.5\, \log \boldsymbol{\sigma}^2\right) \odot \epsilon, \qquad \epsilon \sim \mathcal{N}(0, I). \tag{13}$$

Multiplication in elements: $(\boldsymbol{\sigma} \odot \varepsilon)_i = \boldsymbol{\sigma}_i \varepsilon_i$, where $\odot$ denotes the Hadamard product (in elements).

c. Convolutional decoder and cross-scale skip connections. In the decoding stage, the latent variable z is first projected back to the spatial shape of the last encoder layer. The projected feature map $\widetilde{H}^{(4)}$ is obtained as

$$\widetilde{H}^{(4)} = \mathrm{reshape}(W_d z + b_d) \in \mathbb{R}^{256\times8\times8}, \tag{14}$$

where W_d denotes the projection matrix that maps the 1024-dimensional latent vector z to a tensor of size $256 \times 8 \times 8$, and b_d is the corresponding bias term.

Subsequently, four transposed-convolution blocks (upsampling) are applied to progressively restore spatial resolution along the path $8 \to 16 \to 32 \to 64 \to 128$. To retain high-resolution details, the output of each decoder stage is concatenated with the encoder feature map at the corresponding scale along the channel dimension. Formally, for the lth decoding stage, there is

$$H_{\mathrm{dec}}^{(l)} = \mathrm{DecoderUp}^{(l)}(\mathrm{Concat}(H_{\mathrm{dec}}^{(l+1)}, H_{\mathrm{enc}}^{(l)})), \tag{15}$$

where $H_{\mathrm{dec}}^{(l)}$ is the feature map produced by the lth decoder block, $H_{\mathrm{dec}}^{(l+1)}$ and $H_{\mathrm{enc}}^{(l)}$ denote the feature maps from the $(l+1)$th decoder block and the lth encoder block, respectively, $\mathrm{Concat}(\cdot)$ denotes channel-wise feature concatenation, and $\mathrm{DecoderUp}^{(l)}(\cdot)$ represents the lth transposed-convolution upsampling module. This design allows the reconstruction process to combine global structural information with localized spectral details, as illustrated schematically in Fig. 2.

10 October 2026 10:37:30

The final reconstructed Mel-spectrogram $\widetilde{M}$ is obtained from the output of the first decoder layer via a point-wise nonlinearity,

$$\widetilde{M} = \sigma(H_{\text{dec}}^{(1)}) \in [0,1]^{1\times 128\times 128}, \tag{16}$$

where $\sigma(\cdot)$ denotes the activation function (sigmoid in our implementation), which constrains the decoder output to the range [0, 1] and makes it suitable for subsequent waveform reconstruction.

d. Objective function. The LR-VAE is optimized using a combination of a reconstruction loss and a Kullback–Leibler (KL) divergence regularization term. The loss of reconstruction $\mathcal{L}_{\text{rec}}$ measures the discrepancy between the reconstructed Mel-spectrogram $\hat{M}$ and the reference spectrogram M,

$$\mathcal{L}_{\text{rec}} = \sum_{i,m} (\hat{M}(i,m) - M(i,m))^2. \tag{17}$$

The KL divergence term $\mathcal{L}_{\text{KL}}$ encourages the approximate posterior to remain close to the standard normal prior

$$\mathcal{L}_{\text{KL}} = -\frac{1}{2}\sum_{j=1}^{1024}(1 + \log\sigma_j^2 - \mu_j^2 - \sigma_j^2), \tag{18}$$

where μ_j and σ_j^2 denote the jth mean and variance of the latent variable distribution, respectively.

The overall loss is expressed as

$$\mathcal{L} = \mathcal{L}_{\text{rec}} + 0.1\,\mathcal{L}_{\text{KL}}, \tag{19}$$

where the coefficient 0.1 controls the relative contribution of the KL term.

During training, a gradient-based optimizer updates network parameters in a mini-batch manner, while monitoring the validation loss to determine early stopping. Training is terminated when the validation objective does not improve for several consecutive epochs, and the model checkpoint with the lowest validation loss is selected as the final LR-VAE model.

3. Log-Mel post-processing

Because the LR-VAE output is a normalized Mel-spectrogram $\hat{\boldsymbol{M}} \in \mathbb{R}^{128\times 128}$, additional steps are required to recover physically interpretable magnitudes.

a. Inverse log dynamic-range restoration. Using the stored minimum and maximum log-magnitude values $(M_{\min}, M_{\max})$, the restored log-magnitudes are computed as

$$\widetilde{M}(i,m) = \hat{M}(i,m)(M_{\max} - M_{\min}) + M_{\min}. \tag{20}$$

Mel-band energies are then estimated via

$$E(i,m) = \exp\left(\widetilde{M}(i,m)\right) - \exp(\varepsilon). \tag{21}$$

b. Inverse Mel mapping. The linear-frequency magnitude spectrum is reconstructed as

$$\widetilde{P}(m,k) = \sum_{i=0}^{F-1} E(i,m)H_i(k). \tag{22}$$

Then, the amplitude spectrogram is obtained by

$$\widetilde{A}(m,k) = \sqrt{\widetilde{P}(m,k)}. \tag{23}$$

c. Griffin–Lim phase reconstruction. Since the LR-VAE does not estimate phase information, the Griffin–Lim algorithm is employed to iteratively refine the phase during waveform reconstruction. Let $\phi^{(t)}(m,k)$ denote the phase estimate at the tth iteration. The complex spectrogram $\widetilde{S}^{(t)}(m,k)$ is constructed as

$$\widetilde{S}^{(t)}(m,k) = \widetilde{A}(m,k)\, e^{j\phi^{(t)}(m,k)}, \tag{24}$$

where $\widetilde{A}(m,k)$ is the reconstructed magnitude spectrum. An inverse short-time Fourier transform (ISTFT) is then applied to obtain the corresponding time-domain signal,

$$\widetilde{x}^{(t)}[n] = \text{ISTFT}\left(\widetilde{S}^{(t)}(m,k)\right). \tag{25}$$

A new phase estimate is acquired by performing the STFT on the reconstructed signal,

$$\phi^{(t+1)}(m,k) = \arg\left(\text{STFT}\left(\widetilde{x}^{(t)}[n]\right)\right). \tag{26}$$

After approximately 50–60 iterations, the process typically reaches a steady solution, yielding the final reconstructed waveform,

$$\widetilde{x}[n] = \widetilde{x}^{(T)}[n]. \tag{27}$$

These reconstructed waveforms are subsequently incorporated into the training set as augmented samples. By applying the previously-noted reconstruction procedure to all spectrogram segments from each class, the resulting audio retains the general scale of the original data while introducing additional variability through latent-space perturbations. This diversity helps enrich intra-class distributions and provides more informative training data for the downstream classifier.

D. AIM model

After the ACcut module completes effective-segment selection and the LR-VAE module finishes data augmentation and reconstruction, an AIM model is designed to perform robust feature representation and classification. AIM consists of a BEATs acoustic representation network and a LightGBM structured classifier. Its core motivation is to couple deep representation learning with the strong discriminative capability of gradient-boosted decision trees, thereby

10 October 2026 10:37:30

https://doi.org/10.1121/10.0043727

achieving robust, generalizable, and interpretable recognition performance under few-shot learning conditions, weak inter-class separability, and pronounced noise interference common in underwater acoustic environments.

The general workflow of AIM is as follows. After structural augmentation, all reconstructed segments are fed into a frozen BEATs model to extract 768-dimensional acoustic embeddings. The BEATs network, trained on large-scale speech and acoustic corpora, provides stable and informative frequency–temporal representations even without fine-tuning on underwater data. These embeddings retain both broadband spectral structure and fine-grained temporal patterns. Subsequently, a LightGBM classifier is trained using fivefold stratified cross-validation across the 12 vessel classes. Due to its second-order gradient optimization and direction-aware node splitting, LightGBM is particularly suitable for high-dimensional embedding spaces and offers better resistance to overfitting than fully connected classification heads, especially under limited-data conditions. Moreover, its tree-based decision boundary modeling enhances sensitivity to subtle inter-class differences. Finally, a multi-class Softmax layer produces class posterior probabilities, yielding the final vessel identity prediction.

1. BEATs acoustic representation

BEATs is a large-scale self-supervised acoustic model trained with the Masked Audio Modeling objective. Through multi-layer Transformer encoders and a patch-level representation structure, BEATs provides robust and transferable audio embeddings across diverse acoustic domains. In this work, BEATs is employed to extract high-level acoustic features from LR-VAE reconstructed segments, after preprocessing via the ACcut module.

a. Log-Mel spectrogram extraction. The input waveform is first segmented and windowed using the same parameters as described previously. Let the window length be $N = 1024$ and the hop size be $H = 512$. The mth analysis frame is expressed as

$$x_m[n] = \tilde{x}[n + mH]\, w[n], \quad 0 \le n < N, \tag{28}$$

where $\tilde{x}[n]$ denotes the ACcut-processed waveform and $w[n]$ is the Hann analysis window.

A discrete Fourier transform (DFT) is applied to each frame, yielding the complex spectrum,

$$X_m(k) = \sum_{n=0}^{N-1} x_m[n]\, e^{-j2\pi nk/N}. \tag{29}$$

All spectral frames are then concatenated to form the two-dimensional time–frequency magnitude matrix,

$$S(m,k) = |X_m(k)|^2. \tag{30}$$

The Mel-filterbank is subsequently applied to compress the linear-frequency spectrum into perceptually motivated frequency bands. The Mel-filtered energy for the ith Mel band is given by

$$M(i,m) = \log\left(\sum_{k=0}^{K-1} S(m,k)\, H_i(k) + \varepsilon\right), \tag{31}$$

where $H_i(k)$ denotes the ith triangular Mel filter and ε is a small constant included to avoid numerical instability. For consistency with the LR-VAE input representation, the log-Mel spectrogram is temporally normalized to a fixed size of 128×128 using the same padding and uniform temporal sampling strategy described previously, and is then used as the input feature representation for the BEATs encoder.

b. Patch embedding. The normalized Mel-spectrogram is divided into non-overlapping 16×16 patches. Each patch is linearly projected into a $D = 768$-dimensional embedding

$$z = \mathrm{PatchEmbed}(M) + E_{\mathrm{pos}}, \tag{32}$$

where E_{pos} denotes learnable positional embeddings.

c. Transformer encoder. The patch sequence is fed into a standard $L = 12$-layer Transformer encoder, where each layer consists of a multi-head self-attention (MHSA) module followed by a feed-forward network (FFN). The output of layer l is the following:

$$z_l = \mathrm{TransformerBlock}_l(z_{l-1}), \quad l = 1, 2, \ldots, L. \tag{33}$$

After the final encoder layer, a temporal average pooling is applied to obtain the BEATs-level embedding,

$$\boldsymbol{f}_{\mathrm{BEATs}} = \mathrm{AvgPool}(z_L) \in \mathbb{R}^{768}, \tag{34}$$

where $\mathrm{AvgPool}(\cdot)$ performs patch-wise temporal grouping. BEATs are pretrained on large-scale audio corpora and can extract high-level acoustic representations even in unseen underwater environments.

2. LightGBM-based classification

After obtaining the 768-dimensional deep acoustic embedding $\boldsymbol{f} \in \mathbb{R}^{768}$ from the BEATs encoder, the proposed framework employs LightGBM to perform multi-class classification. LightGBM integrates gradient enhancement with an efficient leaf-wise growth strategy, yielding stable performance even when training data are limited and the acoustic environment is complex.

LightGBM assumes that the final classifier consists of K boosted decision trees. The prediction function for a given feature vector $\boldsymbol{f}$ is defined as

$$F_K(\boldsymbol{f}) = \sum_{k=1}^{K} \eta\, T_k(\boldsymbol{f}), \quad \eta = 0.1, \tag{35}$$

where $T_k(\cdot)$ denotes the kth decision tree and η is the learning rate. The resulting output $F_K(\boldsymbol{f})$ represents the raw score for each class.

10 October 2026 10:37:30

For each BEATs embedding $\boldsymbol{f}_i$, LightGBM outputs a set of raw scores $F^{(c)}(\boldsymbol{f}_i)$ for all classes. These scores are converted to normalized probabilities using the Softmax function,

$$p_{i,c} = \frac{\exp\left(F^{(c)}(\boldsymbol{f}_i)\right)}{\sum_{j=1}^{C} \exp\left(F^{(j)}(\boldsymbol{f}_i)\right)}. \tag{36}$$

The predicted class label for sample i is determined as

$$\hat{y}_i = \arg\max_c \, p_{i,c}, \tag{37}$$

where $\hat{y}_i$ denotes the final predicted category. LightGBM employs the multi-class cross-entropy loss as its optimization target

$$\mathcal{L} = -\sum_{i=1}^{N}\sum_{c=1}^{C} y_{i,c} \log p_{i,c}, \tag{38}$$

where $y_{i,c} \in \{0, 1\}$ represents the one-hot encoded ground-truth label for class c.

Rather than directly minimizing the loss $\mathcal{L}$, LightGBM computes first- and second-order gradients for each sample to update the tree parameters, which are respectively given as

$$g_{i,c} = \frac{\partial \mathcal{L}}{\partial F^{(c)}(\boldsymbol{f}_i)} = p_{i,c} - y_{i,c}, \tag{39}$$

$$h_{i,c} = p_{i,c}\,(1 - p_{i,c}). \tag{40}$$

These gradients are computed directly during the LightGBM training procedure and are consistent with the Softmax-based multi-class output used by the BEATs feature embeddings. LightGBM leverages both gradient orders to construct efficient leaf-wise splits and optimize decision boundaries in a data-efficient manner.

During the training of each decision tree, LightGBM selects the optimal split point based on the following gain formulation:

$$\text{Gain}(S) = \frac{1}{2}\left[\frac{\left(\sum_{i\in S_L} g_i\right)^2}{\sum_{i\in S_L} h_i + \lambda} + \frac{\left(\sum_{i\in S_R} g_i\right)^2}{\sum_{i\in S_R} h_i + \lambda} - \frac{\left(\sum_{i\in S} g_i\right)^2}{\sum_{i\in S} h_i + \lambda}\right] - \gamma, \tag{41}$$

where λ is the L2 regularization coefficient applied to the leaf weights, and γ denotes the minimum split gain threshold used to control the complexity of the tree (the default value is typically zero). This criterion encourages splits that provide substantial improvement in separating the gradient statistics while regularizing the model capacity.

For each leaf node, the optimal output value is computed as

$$\omega = -\frac{\sum_{i\in S} g_i}{\sum_{i\in S} h_i + \lambda}. \tag{42}$$

This leaf weight is then incorporated into the additive boosting model,

$$F_k(\boldsymbol{f}) \leftarrow F_{k-1}(\boldsymbol{f}) + \eta\,\omega, \tag{43}$$

where η represents the learning rate that modulates the contribution of each tree to the final prediction.

In general, the AIM module integrates the BEATs embeddings with the structural advantages of the LightGBM gradient-boosted decision trees. This hybrid architecture provides stable multi-class decisions under few-shot learning and noise-corrupted conditions, offering a strong foundation for the downstream classification tasks.

In summary, the proposed AIM module integrates deep acoustic embeddings from BEATs with the structured second-order optimization capability of LightGBM. This hybrid design enables highly discriminative and noise-robust representation learning under few-shot learning conditions, achieving reliable multi-class ship identification. The model not only enhances generalizability against underwater noise perturbations but also provides a solid decision-making foundation, as verified in subsequent experiments.

III. EXPERIMENTAL SETUP

This chapter provides a systematic description of the datasets, task construction, input feature representation, model training configuration, and baseline systems used in this study. All methods are trained and evaluated under identical preprocessing and data partitioning settings to ensure fair and consistent comparison between models. The experiments were carried out on the basis of PyTorch-2.1 on a platform with an NVIDIA RTX-3090 GPU (NVIDIA, Santa Clara, CA).

A. Datasets

The experiments are conducted on two real-world underwater acoustic datasets: Shipsear[28] and DeepShip.[29] These datasets differ significantly in recording platforms, ambient noise structures, number of classes, and sample scales, allowing evaluation of model generalization under few-shot, multi-class, and noise-varying conditions.

The Shipsear dataset consists of multiple categories of nearshore ship-radiated noise. The recordings are affected by wind–wave interactions, ocean currents, mechanical background noise, and distance attenuation, leading to strong non-stationarity, complex noise, and subtle inter-class spectral differences. In this paper, we use all 12 ship classes in Shipsear. The dataset is first partitioned at the original recording (file) level, such that all segments

10 October 2026 10:37:30

https://doi.org/10.1121/10.0043727

extracted from the same raw audio recording are assigned exclusively to either the training set or the test set, ensuring strict independence between the two sets. After this recording-level split, the ACcut module is applied independently to the training and test recordings to generate valid 5-s waveform segments.

For the Shipsear dataset, categories with fewer than 500 samples are considered small-sample classes, and data augmentation is performed by generating additional samples for these classes, doubling their original size. Classes with more than 500 samples remain unchanged. Furthermore, only 500 samples are selected for each class for training and testing, ensuring consistency throughout the experiment.

The DeepShip dataset is recorded using far-field array systems and exhibits higher signal-to-noise ratios and more regular spectral patterns. Four vessel categories—*Cargo*, *PassengerShip*, *Tanker*, and *Tug*—are selected in this study. Similarly, the DeepShip dataset is first split at the original recording level prior to segmentation, so that no 5-s segments originating from the same recording appear in both the training and test sets. All recordings are then preprocessed using the ACcut module after the file-level split.

For the DeepShip dataset, the *Tug* class is considered a small-sample class with fewer than 1000 samples and is augmented by generating additional samples to double its original size. The other classes, with more than 1000 samples, remain unchanged. Additionally, only 1000 samples per class are selected for training and testing to ensure uniformity across the datasets.

Tables I and Table II summarize the number of samples for each class in the Shipsear and DeepShip datasets before and after VAE augmentation. For both datasets, the final 80%/20% training and test split is applied at the recording level rather than at the segment level, ensuring that no segments derived from the same original audio recording are shared between the two sets. The same partitioning policy is applied to all models to ensure consistency between evaluations.

B. Experimental configuration

A unified experimental task is constructed based on the valid segments obtained from the ACcut preprocessing pipeline, aiming to reflect the overall classification performance under few-shot conditions. First, all recordings are processed by ACcut for cleaning, slicing, and amplitude normalization, producing valid 5-s waveform segments. Next, both Shipsear and DeepShip adopt the same data splitting strategy with strict class balance. Then, all models (VAE-BL, ECAPA-TDNN, CAM++, ResNet-SE, and a Transformer-based classifier) use exactly the same training/testing subsets. Finally, the evaluation metrics include Accuracy, Macro-F1, and equal error rate (EER).

It is important to note that although the original recordings are segmented into 5-s waveform clips, all downstream processing and classification are performed on fixed-dimensional log-Mel spectrogram representations. In particular, LR-VAE-based data augmentation operates at the feature level rather than the waveform level. The generated samples correspond to 128×128 log-Mel representations that summarize the temporal characteristics of the same 5-s segments, rather than shorter waveform fragments. As a result, the AIM classifier, which consumes fixed-length feature embeddings, treats both original and augmented samples as dimensionally and statistically equivalent inputs, without explicit dependence on raw waveform duration.

Based on this unified experimental setup, an additional few-shot evaluation is conducted to further examine the robustness of the proposed method under reduced training data conditions. Specifically, a fixed test set is maintained, while the number of training samples per class is gradually reduced. Throughout this evaluation, all experimental settings, model architectures, and hyperparameters are kept identical to those described previously. This design ensures that the observed performance variations are solely attributed to the reduction in training samples, rather than differences in experimental configuration.

To provide a comprehensive and fair comparison, several representative end-to-end neural classification architectures are selected as baseline systems, including ECAPA-TDNN,[30] CAM++,[31] ResNet-SE,[32] and a Transformer-based classifier.[17] All baseline systems are trained using identical input features, data partitions, and evaluation protocols to ensure a fair and controlled comparison.

TABLE I. Shipsear dataset: Number of samples Before and After VAE Augmentation (80% / 20% split). Total samples and final training/test set sizes are included.

Class	Before VAE	After VAE	Training (80%)	Test (20%)
Dredger	332	664	400	100
Fishboat	443	886	400	100
Motorboat	1902	1902	400	100
MusselBoat	621	621	400	100
NaturalAmbientNoise	936	936	400	100
OceanLiner	442	884	400	100
Passengers	3541	3541	400	100
PilotShip	256	512	400	100
RORO	320	640	400	100
Sailboat	526	526	400	100
Trawler	1110	1110	400	100
Tugboat	257	514	400	100
Total	10 686	12 736	4800	1200

TABLE II. DeepShip Dataset: Number of samples Before and After VAE Augmentation (80% / 20% split). Total samples and final training/test set sizes are included.

Class	Before VAE	After VAE	Training (80%)	Test (20%)
Cargo	2276	2276	800	200
Passengership	1066	1066	800	200
Tanker	1548	1548	800	200
Tug	581	1162	800	200
Total	5471	6052	3200	800

10 October 2026 10:37:30

1. ECAPA-TDNN

ECAPA-TDNN is a state-of-the-art end-to-end speaker recognition architecture that has demonstrated strong performance on large-scale benchmarks such as VoxCeleb. Owing to its channel-attentive time-delay structure and attentive statistical pooling, ECAPA-TDNN serves as a strong reference model for evaluating representation robustness under limited training data and noisy acoustic conditions.

2. CAM++

CAM++ is a lightweight CNN designed for robust acoustic modeling under strong background noise and real-time constraints. By incorporating multi-granularity pooling and compact convolutional blocks, CAM++ exhibits good noise robustness and computational efficiency, making it a suitable baseline for underwater acoustic signal classification.

3. ResNet-SE

ResNet-SE is a residual CNN augmented with squeeze-and-excitation (SE) attention modules. The SE mechanism adaptively recalibrates channel-wise feature responses, enabling the network to emphasize informative spectral patterns while suppressing irrelevant noise components. As a widely adopted CNN backbone, ResNet-SE provides a strong convolutional baseline for evaluating deep feature extraction capability in underwater acoustic environments.

4. Transformer-based classifier

The Transformer-based classifier adopts a self-attention encoder architecture to model long-range temporal dependencies in acoustic feature sequences. By directly operating on log-Mel spectrogram inputs and employing global temporal pooling for classification, the Transformer baseline offers a complementary comparison to convolutional architectures, highlighting the effectiveness of attention-based temporal modeling for ship-radiated noise recognition.

C. Experimental results

This section presents a comprehensive evaluation of the proposed VAE-BL framework on two real-world underwater acoustic datasets, Shipsear and DeepShip. The performance of VAE-BL is compared against representative end-to-end neural baselines, including ECAPA-TDNN,[30] CAM++,[31] ResNet-SE,[32] and a Transformer-based classifier.[17] The analysis covers three aspects: overall classification performance, confusion-matrix-based fine-grained evaluation, and module-level ablation, aiming to validate the effectiveness and generalization ability of the proposed method under multi-noise, multi-class, and sample-scarce conditions.

1. Overall performance comparison

Table III summarizes the overall classification accuracy of five different models on the two datasets. For the Shipsear dataset, VAE-BL achieves an accuracy of **95.56**%, outperforming CAM++ (92.3%), ECAPA-TDNN (90.2%), ResNet-SE (88.3%), and the Transformer-based classifier (85.15%). Given the subtle inter-class spectral differences and the highly non-stationary noise conditions in this 12-class task, VAE-BL demonstrates a stronger ability to suppress background interference and maintain better intra-class compactness and inter-class separability.

In the DeepShip dataset, VAE-BL also achieves the highest accuracy of **90.6**%, compared to CAM++ (88.6%), ECAPA-TDNN (86.3%), ResNet-SE (84.5%), and the Transformer-based classifier (87.24%). In particular, VAE-BL offers substantial improvements for sample-scarce categories such as *Tug*, benefiting from the generative enhancement provided by LR-VAE and the robust acoustic embedding space learned by BEATs-v2.

Overall, VAE-BL consistently achieves the best performance across both datasets, demonstrating strong generalization capabilities under conditions involving few-shot learning, strong noise, and multiple acoustic classes.

2. Confusion matrix analysis

To further investigate the fine-grained classification behavior of different models, this subsection analyzes the confusion matrices obtained on the Shipsear and DeepShip datasets. Confusion matrices provide an intuitive view of the correct classification proportion of each class and reveal major sources of misclassification, thereby enabling evaluation of model robustness under noisy backgrounds, subtle inter-class spectral differences, and imbalanced sample distributions.

a. Shipsear confusion matrices. The Shipsear dataset contains 12 categories of nearshore ship-radiated noise with complex background interference and subtle inter-class variations, making it a challenging multi-class recognition task. Figure 3 presents the Shipsear confusion matrices of five different models, where the horizontal and vertical axes use compact labels C1–C12 for clarity. Specifically, C1–C12 correspond to *Dredger*, *Fishboat*, *Motorboat*, *MusselBoat*, *NaturalAmbientNoise*, *OceanLiner*, *Passengers*, *PilotShip*, *RORO*, *Sailboat*, *Trawler*, and *Tugboat*, respectively. Figures 3(a)–3(d) correspond to CAM++, ECAPA-TDNN, ResNet-SE, and the Transformer-based classifier, respectively, while Fig. 3(e) corresponds to the proposed VAE-BL model.

TABLE III. Overall classification accuracy of different models on Shipsear and DeepShip datasets.

Method	Shipsear Acc (%)	DeepShip Acc (%)
ECAPA-TDNN	90.2	86.3
CAM++	92.3	88.6
ResNet-SE	88.3	84.5
Transformer	85.15	87.24
VAE-BL	**95.56**	**90.6**

10 October 2026 10:37:30

https://doi.org/10.1121/10.0043727

As shown in Fig. 3(e), nearly all diagonal entries exceed 0.90, with classes such as *MusselBoat*, *NaturalAmbientNoise*, and *PilotShip* achieving values above 0.95. Off-diagonal values rarely exceed 0.03, indicating that VAE-BL effectively suppresses spectral perturbations caused by strong ambient noise and produces well-separated latent representations across classes.

Figure 3(a) reveals noticeable confusion between *Motorboat* and *Fishboat*/*MusselBoat*, which can be attributed to the limitations of multi-granularity convolutional modeling in capturing fine spectral details under few-shot conditions. As shown in Fig. 3(b), classes such as *NaturalAmbientNoise* and *OceanLiner* exhibit off-diagonal values of around 0.05, suggesting that ECAPA-TDNN struggles to maintain stable decision boundaries under strong noise and sparse data scenarios.

Figures 3(c) and 3(d) show that ResNet-SE and the Transformer-based classifier exhibit distinct class-wise behaviors. ResNet-SE benefits from channel-wise attention in emphasizing dominant spectral patterns but remains sensitive to spectrally similar vessel types, while the Transformer model leverages global temporal attention to improve discrimination for certain classes, at the cost of reduced robustness under heavily noise-corrupted conditions.

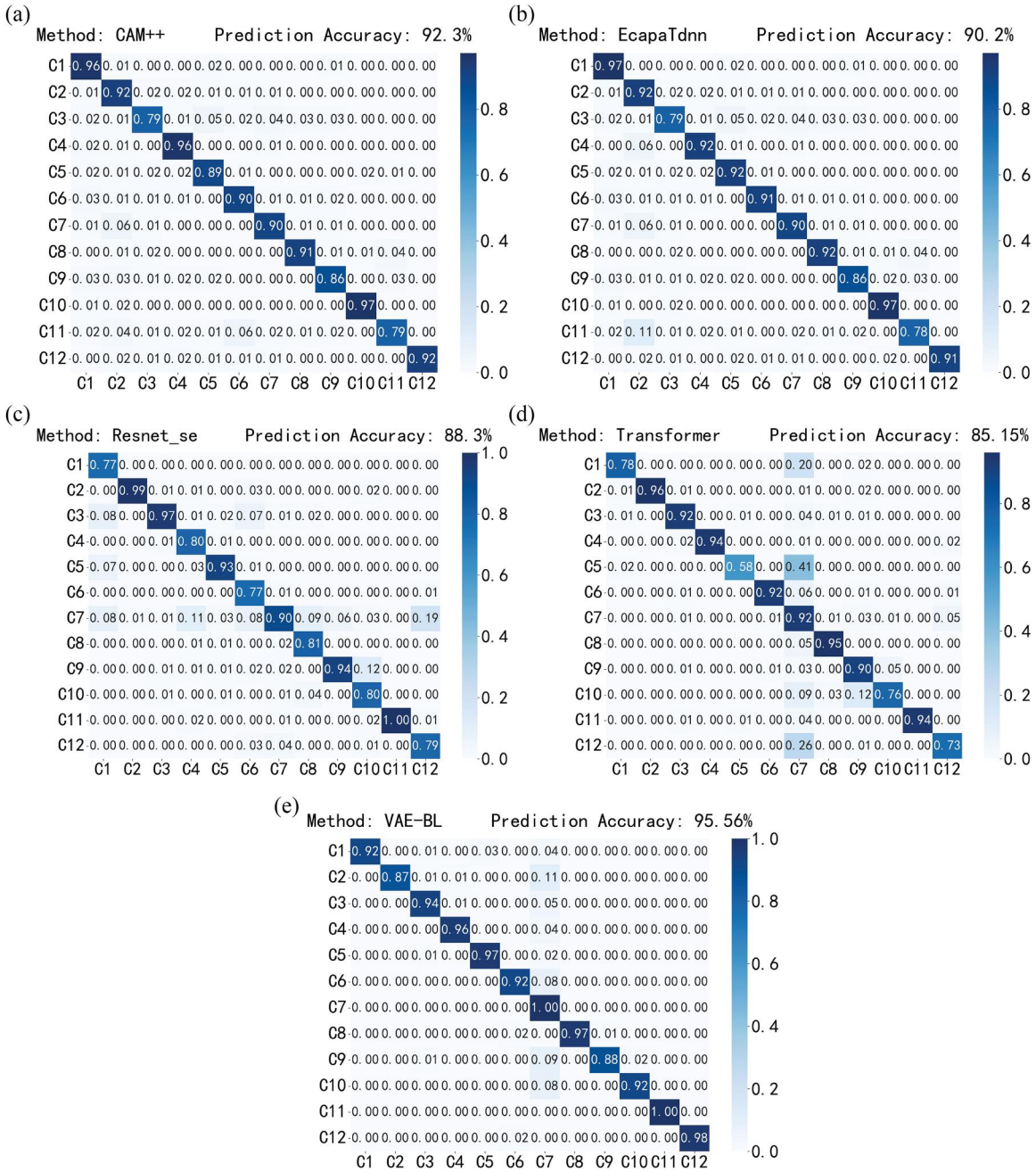


FIG. 3. Shipsear confusion matrices of (a) CAM++, (b) ECAPA-TDNN, (c) ResNet-SE, (d) Transformer-based classifier, and (e) the proposed VAE-BL model.

10 October 2026 10:37:30

TABLE IV. Per-class accuracy of CAM++, ECAPA-TDNN, ResNet-SE, Transformer-based classifier, and VAE-BL on the Shipsear dataset.

Class	CAM++	ECAPA-TDNN	ResNet-SE	Transformer	VAE-BL
Dredger	0.96	0.97	0.77	0.78	0.93
Fishboat	0.92	0.92	0.99	0.96	0.87
Motorboat	0.79	0.79	0.97	0.92	0.93
MusselBoat	0.96	0.92	0.80	0.94	0.95
NaturalNoise	0.89	0.92	0.93	0.58	0.97
OceanLiner	0.90	0.91	0.77	0.92	0.91
Passengers	0.90	0.90	0.90	0.92	0.98
PilotShip	0.91	0.92	0.81	0.95	0.98
RORO	0.86	0.86	0.94	0.90	0.89
Sailboat	0.97	0.97	0.80	0.76	0.91
Trawler	0.79	0.78	1.00	0.94	1.00
Tugboat	0.92	0.91	0.79	0.73	0.98

Table IV reports the per-class classification accuracy of all five models on the Shipsear dataset. Per-class accuracy is computed as the diagonal entry of the row-normalized confusion matrix shown in Fig. 3.

Overall, VAE-BL demonstrates significantly better class separation in multi-noise, multi-class scenarios, especially for spectrally similar vessel types where its confusion regions are markedly smaller than those of the baseline models.

b. DeepShip confusion matrices. The DeepShip dataset contains four categories of far-field ship-radiated noise with an imbalanced class distribution, where the *Tug* class is notably sample-scarce. In Fig. 4, the horizontal and vertical axes adopt compact labels C1–C4 for clarity. Specifically, C1–C4 correspond to *Cargo*, *PassengerShip*, *Tanker*, and *Tug*, respectively. Figures 4(a)–4(d) present the confusion matrices of CAM++, ECAPA-TDNN, ResNet-SE, and the Transformer-based classifier, respectively, while Fig. 4(e) corresponds to the proposed VAE-BL model.

As illustrated in Fig. 4(e), the accuracies per-class fall within the range of 0.85–0.95, with the *Tug* category reaching 0.95, substantially outperforming the two end-to-end baselines. The results in Table V further confirm the superiority of VAE-BL in sample-sparce categories. Figure 4(a) shows an occasional misclassification between *PassengerShip* and *Tanker*, although the overall distribution remains balanced. In Fig. 4(b), there is a notable confusion between *Tanker* and *Tug*, with the lowest diagonal value at 0.83. This is consistent with the model's reliance on abundant training samples to maintain separability in the embedding space. Table V summarizes the accuracy per-class of the three models (CAM++, ECAPA-TDNN, and VAE-BL) in the DeepShip dataset.

In summary, the DeepShip experiments further validate the robustness of VAE-BL under imbalanced and sample-scarce settings. Together with the results in Table V, it can be concluded that the performance gains of VAE-BL primarily result from the improved spectral consistency introduced by LR-VAE reconstruction and the stable representational capacity of BEATs-v2 embeddings.

3. Few-shot performance with reduced training samples

To further evaluate the robustness of the proposed VAE-BL framework under more challenging few-shot conditions, we conduct an additional set of experiments by gradually reducing the number of training samples per class while keeping the test sets fixed. This evaluation is performed on both Shipsear and DeepShip datasets based on the unified experimental configuration described previously.

For the Shipsear dataset, the number of training samples per class is reduced from the original setting of 400 to 300, 200, 100, and 50, while the test set remains unchanged. As reported in Table VI, the classification precision gradually decreases as the training data become more limited. Starting from a baseline accuracy of 95.56%, the performance remains relatively stable at 95.14% and 94.76% when the training samples are reduced to 300 and 200 per class, respectively. When the training size is further reduced to 100 and 50 samples per class, the precision drops to 93.12% and 90.48%. Despite the reduction, the proposed method maintains strong recognition performance even under extremely limited training data, demonstrating its robustness in few-shot scenarios.

A similar trend is observed on the DeepShip dataset. The number of training samples per class is reduced from 800 to 600, 400, 200, and 100, while the test set is kept fixed. The accuracy decreases from the baseline of 90.6% to 90.2%, 89.6%, 88.1%, and 85.5%, respectively. Compared with results in the full training setting, the performance degradation remains gradual, indicating that VAE-BL exhibits stable generalization capability when training data is scarce.

In general, the results in Table VI confirm that the proposed method achieves a relatively slow degradation in performance as the number of training samples decreases. This behavior highlights the effectiveness of LR-VAE-based generative augmentation and the robust acoustic representations learned by BEATs-v2, which together enable reliable recognition under limited training data conditions.

4. Ablation study

To thoroughly assess the individual contribution of each component within the proposed VAE-BL framework,

10 October 2026 10:37:30

https://doi.org/10.1121/10.0043727

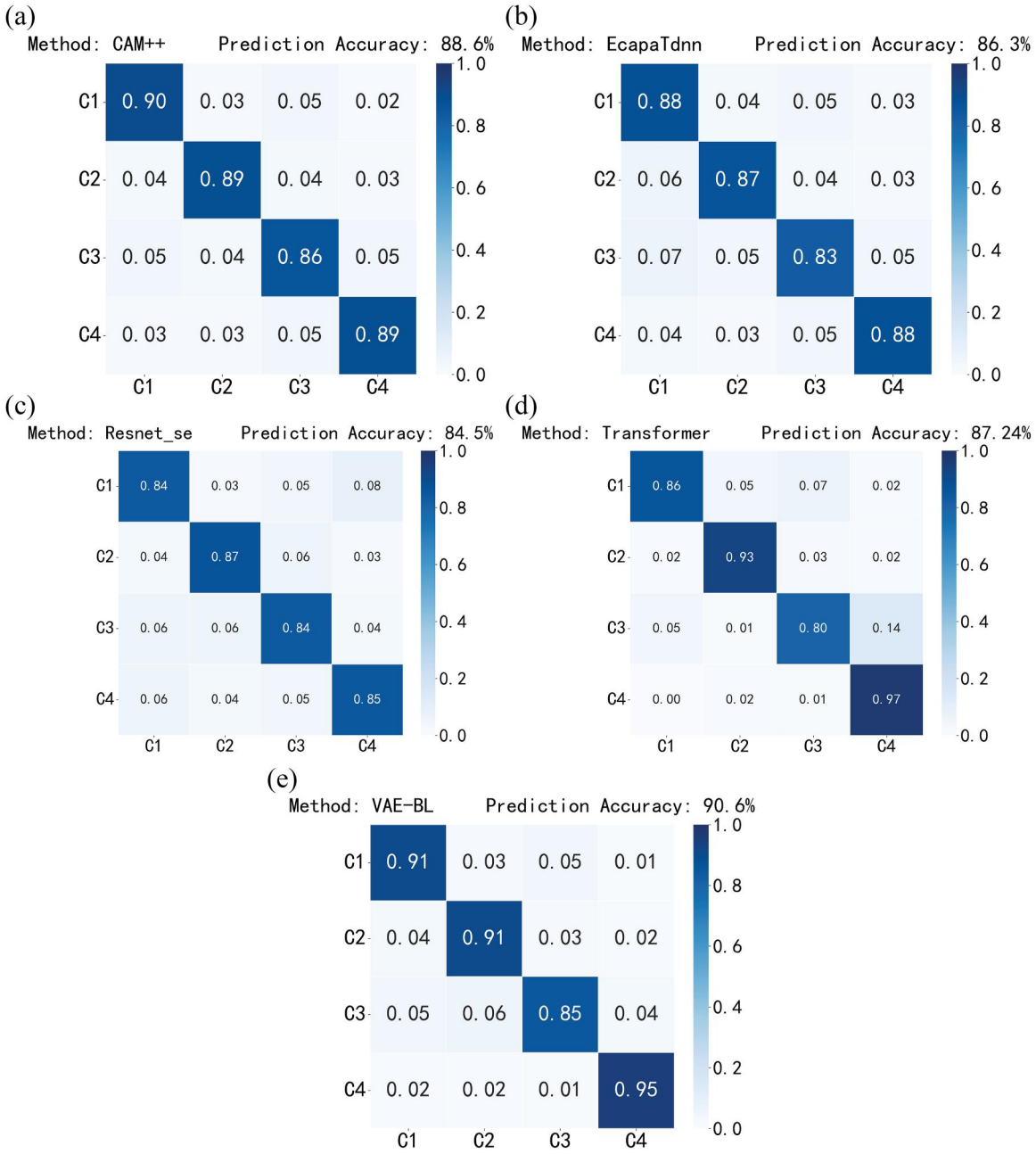


FIG. 4. DeepShip confusion matrices of (a) CAM++, (b) ECAPA-TDNN, (c) ResNet-SE, (d) Transformer-based classifier, and (e) the proposed VAE-BL model.

we perform a series of ablation studies on the ShipsEar and DeepShip datasets. The analysis is conducted at three levels: (i) removing the core modules (LR-VAE, BEATs-v2, AIM), (ii) evaluating the architectural variations within LR-VAE, and (iii) studying the impact of different augmentation ratios. These experiments provide a comprehensive understanding of how each component contributes to the robustness and discriminative capability of the whole system.

TABLE V. Per-class accuracy of CAM++, ECAPA-TDNN, ResNet-SE, Transformer-based classifier, and VAE-BL on the DeepShip dataset.

Class	CAM++	ECAPA-TDNN	ResNet-SE	Transformer	VAE-BL
Cargo	0.90	0.88	0.84	0.86	0.91
PassengerShip	0.89	0.87	0.87	0.93	0.91
Tanker	0.86	0.83	0.84	0.80	0.85
Tug	0.89	0.88	0.85	0.97	0.95

10 October 2026 10:37:30

TABLE VI. Few-shot classification accuracy of VAE-BL under reduced training sample sizes on Shipsear and DeepShip datasets. The test sets are kept fixed across all experiments.

Training samples per class	Shipsear Acc (%)	DeepShip Acc (%)
400 / 800	95.56	90.6
300 / 600	95.14	90.2
200 / 400	94.76	89.6
100 / 200	93.12	88.1
50 / 100	90.48	85.5

a. Effect of removing individual modules. Figure 5 illustrates the variations in classification accuracy and Macro-F1 in different modules removed. The baseline configuration (VAE-BL-Full) incorporates all functional modules, including ACcut preprocessing, LR-VAE reconstruction with BEATs-v2 embeddings, and the AIM classifier. Three variants are evaluated: (1) *w/o LR-VAE*, where the generative reconstruction module is removed; (2) *w/o BEATs-v2*, where the acoustic embeddings are replaced by raw log-Mel features; and (3) *w/o AIM*, where the LightGBM-based classifier is replaced by a fully connected layer.

Removing LR-VAE. Without LR-VAE, the model accuracy drops by approximately 4%–5% on both datasets, with Macro-F1 showing a similar notable reduction. This confirms that the reconstruction-based consistency enhancement of LR-VAE effectively reduces intra-class variance under noisy conditions, yielding more compact latent representations.

Removing BEATs-v2. Replacement of BEATs-v2 with conventional log-Mel features results in the most severe degradation, with accuracy reductions exceeding 6% and substantial decreases in Macro-F1. This highlights the strong robustness and discriminative power of BEATs-v2 in few-shot acoustic classification scenarios.

Removing AIM. Substituting AIM with a fully connected classifier leads to moderate performance deterioration. The LightGBM-based AIM classifier demonstrates a superior ability in exploiting the high-dimensional BEATs-v2 feature space and forming stable nonlinear decision boundaries.

b. Effect of LR-VAE architectural components. Figure 6 visualizes the performance of several LR-VAE architectural variants. The analysis examines the impact of multi-scale skip connections, latent dimensionality, and KL divergence weight β.

Key observations are as follows:

- **Skip connections are essential;** removal of them reduces accuracy by 2.63%, confirming their role in preserving high-frequency details during reconstruction.
- **The latent dimensionality exhibits a U-shaped effect,** whereas a dimension of 64 is insufficient to capture spectral variations, whereas 256 dimensions offer marginal improvement at the cost of increased computational complexity. The standard 128-dimensional bottleneck achieves the best balance.
- **The KL weight β controls the trade-off between structure and diversity.** $\beta = 0.5$ yields optimal performance. Lower values (e.g., 0.1) under-regularize the latent space, while higher values (e.g., 1.0) over-constrain it and reduce sample diversity.

c. Effect of data augmentation ratios. Figure 7(a) demonstrates the influence of different augmentation ratios on classification performance, while Fig. 7(b) reports the corresponding computational cost in terms of training time. Performance improves consistently as the augmentation ratio increases from 0% to 100%, with gains of 6.12% in accuracy and 0.063 in Macro-F1. The EER decreases from 18.42% to 11.92%, indicating stronger discriminability. However, when the ratio exceeds 100%, redundant or overly similar synthetic samples introduce diminishing returns and slight degradation. Training time scales approximately linearly with the augmentation ratio, ranging from 1.2 h (0%) to 4.9 h (200%). Overall, the 100% augmentation ratio achieves the optimal balance between performance enhancement and training efficiency.

d. LR-VAE reconstruction quality visualization. It should be noted that the log-Mel spectrograms shown in Fig. 8 are not intended to be perceptually realistic reconstructions of raw underwater acoustic signals. Instead, they represent heavily normalized and temporally resampled feature-level visualizations used for generative augmentation. The apparent smoothing effects and blob-like or checkerboard patterns are mainly attributed to the log-Mel

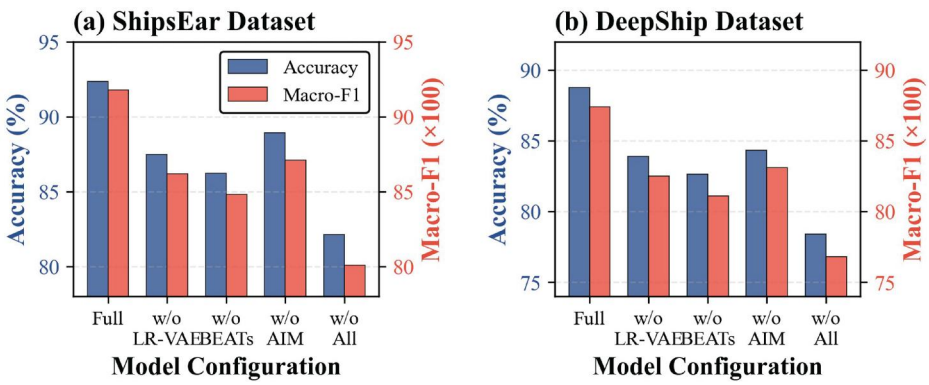


FIG. 5. Performance comparison of VAE-BL and its ablated variants on the ShipsEar and DeepShip datasets. Accuracy and Macro-F1 are reported for the full model and variants without LR-VAE, BEATs-v2, or AIM.

10 October 2026 10:37:30

https://doi.org/10.1121/10.0043727

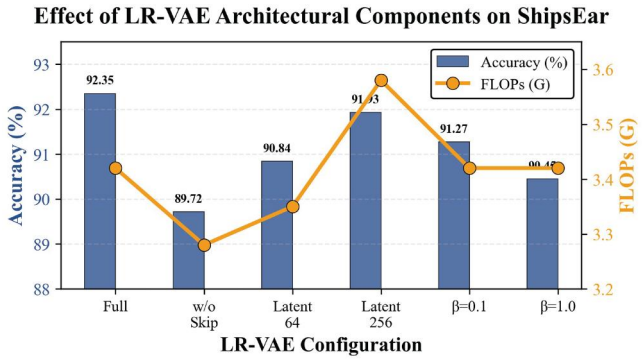


FIG. 6. Ablation of LR-VAE architectural components, including the effects of skip-connection removal, varying latent dimensionality (64/128/256), and changing the KL-divergence weight β. Accuracy and FLOPs are reported on the ShipsEar dataset.

transformation, temporal normalization to a fixed 128-frame resolution, and the use of transposed convolutions in the LR-VAE decoder. Importantly, the objective of LR-VAE is not to synthesize physically faithful waveforms or fine-grained harmonic structures, but to reconstruct and regularize discriminative feature representations in the feature space. As a result, the performance gains do not rely on perceptual realism of the reconstructed spectrograms, but instead stem from improved intra-class consistency and feature space regularization. This interpretation is further supported by the quantitative classification results and the cross-dataset ablation study in Fig. 9, which demonstrate that LR-VAE contributes to performance improvement even though the reconstructed patterns appear visually abstract.

e. Module ablation across datasets. Figure 9 summarizes the contribution of key modules via an ablation study on both ShipsEar and DeepShip. Compared with the Baseline, the full configuration (VAE-BL-Full) consistently improves both Acc and F1 on the two datasets, indicating good cross-dataset robustness. In particular, removing LR-VAE leads to a clear drop in performance, demonstrating that LR-VAE-based reconstruction/augmentation provides substantial gains. Removing BEATs-v2 also results in pronounced degradations, suggesting that the foundation-model encoder is critical for extracting transferable and discriminative representations. The ablation results, therefore, validate that the proposed system benefits from the complementary effects of LR-VAE and BEATs-v2, with AIM providing additional but comparatively smaller improvements.

5. Discussion and summary of findings

The results and confusion-matrix analyzes collectively demonstrate that the proposed VAE-BL framework achieves robust vessel classification across diverse acoustic conditions and few-shot learning scenarios, outperforming representative baselines such as CAM++ and ECAPA-TDNN. The effectiveness of VAE-BL can be attributed to three key mechanisms.

First, the ACcut module ensures reliable input quality by performing fine-grained segmentation and discarding low-energy or noisy fragments. This improves intra-class consistency and stabilizes the downstream learning process, particularly for categories with sparse temporal–spectral features. ACcut also reduces the degree of sample imbalance by filtering excessively heterogeneous segments, improving the reliability of training.

Second, the latent reconstruction mechanism of LR-VAE enhances the separability of categories with similar spectral shapes. Unlike traditional augmentation methods such as SpecAugment—which primarily introduce perturbations in time or frequency—latent reconstruction strengthens category-level invariances and suppresses noise-induced distortions. This is especially beneficial under strong ocean background noise, where the proposed model achieves accuracies of up to 1.00 and 0.98 for several difficult few-shot learning vessel types. In contrast, CAM++ and ECAPA-TDNN exhibit noticeable confusion between certain categories (e.g., CAM++ between *Musselboats* and *Fishboats*, ECAPA-TDNN between *Passenger ships* and *Trawlers*). This confirms that the proposed LR-VAE framework produces more discriminative latent manifolds, leading to well-separated decision boundaries.

In general, these findings show that VAE-BL effectively mitigates the distribution fragmentation and weak discriminability commonly observed in underwater acoustic classification tasks, providing a substantially more robust and generalizable feature representation compared to traditional baselines.

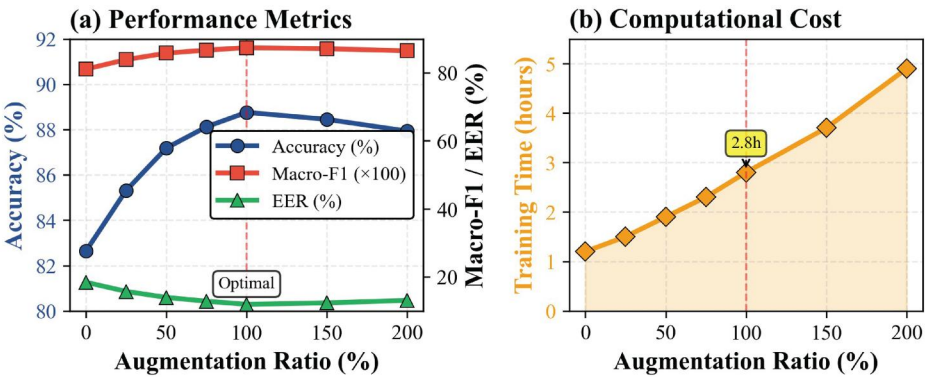


FIG. 7. Effect of LR-VAE augmentation ratios on the DeepShip dataset. (a) Classification performance in terms of Accuracy, Macro-F1, and EER under different augmentation levels. (b) Corresponding computational cost measured by training time.

10 October 2026 10:37:30

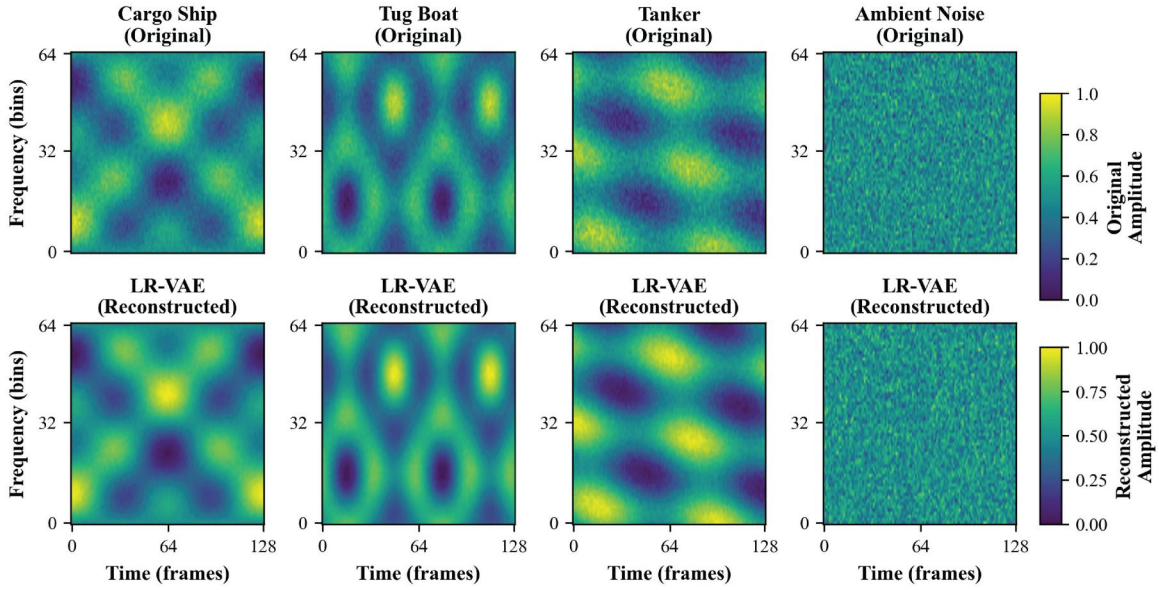


FIG. 8. Visualization of LR-VAE reconstruction quality across different vessel types. Each column compares the original log-Mel spectrogram (top) with the LR-VAE reconstructed version (bottom), illustrating feature-level consistency enhancement.

IV. CONCLUSION

In order to solve the problem of overfitting in small sample object recognition leading to decreased accuracy, and to address the difficulties in designing traditional GAN networks, we propose a VAE-BL framework. LR-VAE adopts a convolutional encoding-decoding architecture with multi-scale skip connections, which preserves local spectral textures and global time-frequency trends in the multi-resolution domain, and maps reparameterized latent variables to structure-preserving latent space representations. Through KL regularization and reconstruction consistency constraints, the latent space remains continuous, differentiable, and has intra-class shrinkage under small sample conditions, thus completing the missing local structures in the original spectrum due to noise or sparse sampling. In the discrimination stage, the framework freezes the BEATs encoder to stabilize the extraction of high-dimensional acoustic features, and then uses LightGBM's leaf node straight-line strategy and gradient splitting criterion to construct a lightweight decision surface. This combination effectively reduces the dependence of deep discriminators on sample size, enabling the model to maintain robust decision-making performance even when there are large differences in feature scales and complex noise scenes. These two parts work together to enable VAE-BL not only to generate continuous intra-class manifolds and enhance data coverage in the latent space, but also to efficiently utilize this completed structure through lightweight discrimination modules, thereby improving intra-class consistency and inter-class separability, ultimately achieving stable recognition performance under extremely few-shot learning and strong noise conditions.

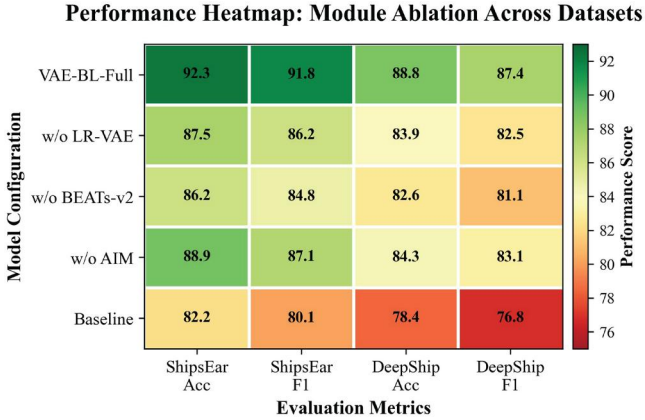


FIG. 9. Performance heatmap of module ablation across the ShipsEar and DeepShip datasets. Rows indicate model configurations (Baseline, w/o AIM, w/o BEATs-v2, w/o LR-VAE, and VAE-BL-Full), and columns report Acc and F1 on both datasets. The full model (VAE-BL-Full) achieves the best overall performance, while removing LR-VAE or BEATs-v2 yields notable degradations, confirming their contributions.

ACKNOWLEDGMENTS

This work was financially supported in part by the National Natural Science Foundation of China (42404001), in part by National Key Research and Development Program of China (2024YFB3909701), in part by the National Natural Science Foundation of China (62271459), in part by Shandong Provincial Natural Science Foundation (ZR2023QF128), and in part by the Fundamental Research Funds for the Central Universities, Ocean University of China under Grant No. 202562022.

AUTHOR DECLARATIONS

Conflict of Interest

The authors have no conflicts to disclose.

10 October 2026 10:37:30

https://doi.org/10.1121/10.0043727

DATA AVAILABILITY

The data that support the findings of this study are available from the corresponding author upon reasonable request.

[1]X. Luo, L. Chen, H. Zhou, and H. Cao, "A survey of underwater acoustic target recognition methods based on machine learning," J. Mech. Sci. Eng. **11**(2), 384 (2023).
[2]Q. Zhang, L. Da, C. Wang, Y. Zhang, and J. Zhuo, "An overview on underwater acoustic passive target recognition based on deep learning," J. Electron. Inf. Technol. **45**(221301), 4190–4202 (2023).
[3]K. Panetta, L. Kezebou, V. Oludare, and S. Agaian, "Comprehensive underwater object tracking benchmark dataset and underwater image enhancement with GAN," IEEE J. Oceanic Eng. **47**(1), 59–75 (2022).
[4]A. Akan and O. Karabiber Cura, "Time–frequency signal processing: Today and future," Digital Signal Process. **119**, 103216 (2021).
[5]V. Pulkki, S. Delikaris-Manias, and A. Politis, *Time–Frequency Processing: Methods and Tools* (IEEE, New York).
[6]S. Imai, "Cepstral analysis synthesis on the Mel frequency scale," in *Proceedings of ICASSP '83. IEEE International Conference on Acoustics, Speech, and Signal Processing* (1983), pp. 93–96.
[7]Z. K. Abdul and A. K. Al-Talabani, "Mel frequency cepstral coefficient and its applications: A review," IEEE Access **10**, 122136–122158 (2022).
[8]N. Tang, F. Zhou, Y. Wang, H. Zhang, T. Lyu, Z. Wang, and L. Chang, "Differential treatment for time and frequency dimensions in Mel-spectrograms: An efficient 3D spectrogram network for underwater acoustic target classification," Ocean Eng. **287**, 115863 (2023).
[9]S. K. Mohammed, S. M. Hariharan, and S. Kamal, "A GTCC-based underwater HMM target classifier with fading channel compensation," J. Sens. **2018**(1), 6593037 (2018).
[10]W. Xue and T. Jiang, "An adaptive algorithm for target recognition using Gaussian mixture models," Measurement **124**, 233–240 (2018).
[11]U. S. Bist and N. Singh, "Analysis of recent advancements in support vector machine," Concurrency Comput. **34**(25), e7270 (2022).
[12]O. Aodha, R. Gibb, K. Barlow, E. Browning, M. Firman, R. Freeman, B. Harder, L. Kinsey, G. Mead, S. Newson, I. Pandourski, S. Parsons, J. Russ, A. Szodoray-Parádi, F. Szodoray-Parádi, E. Tilova, M. Girolami, G. Brostow, and K. Jones, "Bat detective—Deep learning tools for bat acoustic signal detection," PLoS Comput. Biol. **14**, e1005995 (2018).
[13]H. Yang, J. Li, S. Shen, and G. Xu, "A deep convolutional neural network inspired by auditory perception for underwater acoustic target recognition," Sensors **19**(5), 1104 (2019).
[14]Y. Miao, Y. Zakharov, H. Sun, J. Li, and J. Wang, "Underwater acoustic signal classification based on sparse time-frequency representation and deep learning," IEEE J. Oceanic Eng. **46**, 952–962 (2020).
[15]X. Xiao, W. Wang, Q. Ren, P. Gerstoft, and L. Ma, "Underwater acoustic target recognition using attention-based deep neural network," JASA Express Lett. **1**(10), 106001 (2021).
[16]L. Xue, X. Zeng, and A. Jin, "A novel deep-learning method with channel attention mechanism for underwater target recognition," Sensors **22**, 5492 (2022).
[17]S. Feng and X. Zhu, "A transformer-based deep learning network for underwater acoustic target recognition," IEEE Geosci. Remote Sens. Lett. **19**, 1505805 (2022).
[18]A. Thanda and S. M. Venkatesan, "Multi-task learning of deep neural networks for audio visual automatic speech recognition," arXiv:1701.02477 [math-ph] (2017).
[19]Q. Kong, Y. Cao, T. Iqbal, Y. Xu, W. Wang, and M. D. Plumbley, "Cross-task learning for audio tagging, sound event detection and spatial localization: Dcase 2019 baseline systems," arXiv:1904.03476 [math-ph] (2019).
[20]T. Khandelwal and R. K. Das, "A multi-task learning framework for sound event detection using high-level acoustic characteristics of sounds," arXiv:2305.10729 [math-ph] (2023).
[21]F. Liu, T. Shen, Z. Luo, D. Zhao, and S. Guo, "Underwater target recognition using convolutional recurrent neural networks with 3-D Mel-spectrogram and data augmentation," Appl. Acoust. **178**, 107989 (2021).
[22]Z. Jiang, C. Zhao, and H. Wang, "Classification of underwater target based on S-ResNet and modified DCGAN models," Sensors **22**(6), 2293 (2022).
[23]Q. Yao, Y. Wang, and Y. Yang, "Underwater acoustic target recognition based on data augmentation and residual CNN," Electronics **12**(5), 1206 (2023).
[24]K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition* (2016), pp. 770–778.
[25]D. Griffin and J. Lim, "Signal estimation from modified short-time fourier transform," IEEE Trans. Acoust. Speech. Signal Process. **32**(2), 236–243 (1984).
[26]S. Chen, Y. Wu, C. Wang, S. Liu, D. Tompkins, Z. Chen, and F. Wei, "Beats: Audio pre-training with acoustic tokenizers," arXiv:2212.09058 [math-ph] (2022).
[27]G. Ke, Q. Meng, T. Finley, T. Wang, W. Chen, W. Ma, Q. Ye, and T.-Y. Liu, "LightGBM: A highly efficient gradient boosting decision tree," in *Proceedings of Neural Information Processing Systems* (2017).
[28]D. Santos-Domínguez, S. Torres-Guijarro, A. Cardenal-López, and A. Pena-Gimenez, "Shipsear: An underwater vessel noise database," Appl. Acoust. **113**, 64–69 (2016).
[29]M. Irfan, Z. Jiangbin, S. Ali, M. Iqbal, Z. Masood, and U. Hamid, "Deepship: An underwater acoustic benchmark dataset and a separable convolution based autoencoder for classification," Expert Syst. Appl. **183**, 115270 (2021).
[30]X. Du and F. Hong, "Qiandaoear22: A high quality noise dataset for identifying specific ship from multiple underwater acoustic targets using ship-radiated noise," arXiv:2406.04354 [math-ph] (2024).
[31]H. Wang, S. Z. andYafeng Chen, L. Cheng, and Q. Chen, "CAM++: A fast and efficient network for speaker verification using context-aware masking," arXiv:2303.00332 [math-ph] (2023).
[32]J. Hu, L. Shen, and G. Sun, "Squeeze-and-excitation networks," in *Proceedings of the 2018 IEEE/CVF Conference on Computer Vision and Pattern Recognition* (2018), pp. 7132–7141.

10 October 2026 10:37:30